Fact & Dimension Identification & Bus Matrix — Courier DW

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1. Executive Summary

In the **Kimball Enterprise Bus Architecture**, declaring business processes, choosing fact table archetypes, and defining conformed dimensions form the foundation of dimensional modeling. A well-designed dimensional architecture decouples operational event recording from business analytical reporting, ensuring that facts represent quantifiable business measurements and dimensions provide rich descriptive context for filtering, grouping, and slicing.

This deliverable formalizes:

- **Fact Table Identification:** Specifications for three distinct Kimball fact table archetypes (**Transaction**, **Periodic Snapshot**, and **Factless**).
- **Dimension Identification & Hierarchies:** Complete definitions of all nine enterprise dimensions, including natural vs. surrogate keys, attribute catalogs, and explicit roll-up hierarchies.
- **Enterprise Dimensional Bus Matrix:** Standardized matrix mapping facts against conformed dimensions to support cross-process drill-across reporting.
- **Star Schema vs. Snowflake Justification:** Rationale for denormalizing dimensional hierarchies into stars rather than snowflaking.
- **Slowly Changing Dimension (SCD) Policies:** Formal tracking policies for evolving business master data.

2. Fact Table Identification & Process Mapping

The LogiTrack warehouse models three core operational processes, utilizing all three classic Kimball fact table archetypes.

Fact Table Name	Fact Archetype	Business Process Modeled	Grain Definition

Fact_Delivery_Attempt	**Transaction Fact**	Last-mile parcel delivery attempts and exception lifecycle.	One row represents a single physical delivery attempt event for a specific parcel package.
Fact_Daily_Hub_Backlog	**Periodic Snapshot Fact**	Daily sorting hub inventory balances and backlog dwell status.	One row represents the cumulative backlog status and dwell metrics for a specific parcel type at a specific hub at 23:59:59 daily.
Fact_Rider_Route_Assignment	**Factless Fact**	Scheduled workforce fleet allocation and geographic route coverage.	One row represents the scheduled deployment assignment of an individual rider to a designated route, hub, shift, and date (keys only).

2.1 Measure Classification by Fact Table

Fact Table	Fully Additive Measures	Semi-Additive Measures	Non-Additive / D

Fact_Delivery_Attempt	`Delivery_Fee_Amount`
	`Estimated_Labor_Cost`
	`Estimated_Fuel_Cost`
	`Net_Margin_Amount`
	`Attempt_Duration_Minutes`
	`Package_Weight_Kg`
		None (All measures are fully additive across all dimensions).	`On_Time_Del`
			`%)`
			`First_Attem`
			`(FADR %)`
			`Profit_Marg`

Fact_Daily_Hub_Backlog	`Daily_Inflow_Count`
	`Daily_Outflow_Count`
	`Total_Dwell_Hours`
		`Backlog_Parcel_Count` (sums across hubs/types; average across dates)
		`Average_Dwell_Hours`
		`Max_Dwell_Hours`
			`Hub_Backlog`
			`Clearance_V`

Fact_Rider_Route_Assignment	*None* (Strict factless fact: contains zero numeric measure columns).	*None*	`Deliveries_`
			`(DPRH)` (Derived
			`Fact_Delive`
			`Dim_Shift`).

3. Dimension Identification, Hierarchies & Attribute Catalog

The warehouse implements **nine enterprise dimensions**. Every dimension utilizes an artificial integer surrogate primary key to decouple the warehouse from operational system key changes.

1. Dim_Date (Conformed Role-Playing Dimension)

Primary Key: `Date_Key` (INT) | **Natural Key:** `Date_Value` (DATE) | **SCD:** Type 0 (Static)
Explicit Hierarchies: Calendar: Day → Month → Quarter → Year | Weekly: Day → Week of Year → Year
Key Attributes: `Date_Key`, `Date_Value`, `Day_Of_Week_Number`, `Day_Of_Week_Name`, `Day_Of_Month`, `Month_Number`, `Month_Name`, `Quarter_Number`, `Quarter_Name`, `Year`, `Year_Month`, `Week_Of_Year_Number`, `Is_Weekend_Flag`, `Is_Working_Day_Flag`, `Statutory_Holiday_Flag`, `Fiscal_Period`.
Conformed Roles: `Attempt_Date_Key`, `Snapshot_Date_Key`, `Assignment_Date_Key`, `SLA_Deadline_Date_Key`.

2. Dim_Customer (Customer Master Dimension)

Primary Key: `Customer_Key` (INT) | **Natural Key:** `Customer_ID` (VARCHAR) | **SCD:** Type 2 (History tracking)
Explicit Hierarchies: Account: Customer → Customer Segment → Contract Tier | Geographic: Customer → City → Region
Key Attributes: `Customer_Key`, `Customer_ID`, `Company_Name`, `Contact_Person`, `Customer_Segment` (B2B Enterprise, B2B SME, B2C Individual), `Contract_Tier` (Gold, Silver, Bronze), `Credit_Limit_Amount`, `Billing_Street_Address`, `City`, `Postal_Code`, `Region_Name`, `Effective_Date`, `Expiration_Date`, `Current_Row_Indicator`.

3. Dim_Rider (Workforce & Fleet Dimension)

Primary Key: `Rider_Key` (INT) | **Natural Key:** `Rider_ID` (VARCHAR) | **SCD:** Type 2 (History tracking)
Explicit Hierarchies: Assignment: Rider → Home Hub → Region | Fleet: Rider → Vehicle Type → Employment Type
Key Attributes: `Rider_Key`, `Rider_ID`, `First_Name`, `Last_Name`, `Mobile_Phone`, `Employment_Type` (Full-Time Salaried, Freelance Contractor), `Vehicle_Type` (Motorcycle, Van, Bicycle), `Vehicle_Registration_No`, `Home_Hub_Code`, `Home_Region_Name`, `Rating_Score`, `Hire_Date`, `Effective_Date`, `Expiration_Date`, `Current_Row_Indicator`.

4. Dim_Hub (Logistics Sorting Facility Dimension)

Primary Key: `Hub_Key` (INT) | **Natural Key:** `Hub_Code` (VARCHAR) | **SCD:** Type 1 (In-place overwrite)
Explicit Hierarchies: Facility: Hub → Hub Classification Type | Geographic: Hub → City → Region → Country
Key Attributes: `Hub_Key`, `Hub_Code`, `Hub_Name`, `Hub_Type` (Super-Hub Sorting Center, Transit Cross-Dock, Local Depot), `Maximum_Daily_Parcel_Capacity`, `Total_Dock_Doors_Count`,

Hub_Manager_Name, Contact_Phone, Street_Address, City, Postal_Code, Region_Name, Country_Name.

5. Dim_Route (Geographic Delivery Corridor Dimension)

Primary Key: Route_Key (INT) | **Natural Key:** Route_Code (VARCHAR) | **SCD:** Type 1 (In-place overwrite)
Explicit Hierarchies: Corridor: Route → Assigned Servicing Hub → City → Region | Density: Route → Route Zone Classification
Key Attributes: Route_Key, Route_Code, Route_Name, Assigned_Hub_Code, Zone_Classification, City, Region_Name (Urban High-Density, Suburban Residential, Commercial CBD, Industrial Park), Target_Daily_Stops, Average_Route_Distance_Km, Risk_Level_Rating.

6. Dim_Service_Type (Commercial Service Offering Dimension)

Primary Key: Service_Type_Key (INT) | **Natural Key:** Service_Code (VARCHAR) | **SCD:** Type 1 (In-place overwrite)
Explicit Hierarchies: Service: Service Tier → Commitment Window (Hours)
Key Attributes: Service_Type_Key, Service_Code, Service_Tier_Name (Same-Day Express, Next-Day Priority, Standard Economy), Guaranteed_Delivery_Hours (e.g., 4, 24, 72 hrs), Base_Flat_Fee, Fuel_Surcharge_Rate_Pct, Weekend_Pickup_Allowed_Flag.

7. Dim_Delivery_Status (Junk Dimension for Delivery Flags & Exception Codes)

Primary Key: Delivery_Status_Key (INT) | **Natural Key:** Status_Composite_Code | **SCD:** Type 0 (Static)
Explicit Hierarchies: Outcome: Attempt Outcome → Status Category → Specific Failure Exception Reason
Key Attributes: Delivery_Status_Key, Attempt_Outcome (Delivered, Delivery Failed, Customer Refused), Attempt_Status_Category (Successful Delivery, Delivery Exception, Return to Origin), Failure_Reason_Code (None, Customer Unavailable, Incorrect Address, Gate / Access Locked, Weather Disruption, Cash Not Ready), Signature_Required_Flag, Proof_Of_Delivery_Type.

8. Dim_Parcel_Type (Package Classification Dimension)

Primary Key: Parcel_Type_Key (INT) | **Natural Key:** Parcel_Type_Code (VARCHAR) | **SCD:** Type 1 (In-place overwrite)
Explicit Hierarchies: Packaging: Parcel Category → Weight Class → Volumetric Bucket
Key Attributes: Parcel_Type_Key, Parcel_Type_Code, Parcel_Category_Name (Document Flyer, Small Parcel, Medium Carton, Heavy Freight, Fragile Glassware), Weight_Class (Light <1kg, Medium 1-5kg, Heavy 5-20kg, Oversized >20kg), Max_Volume_Cm3, Requires_Two_Man_Handling_Flag, Conveyable_Sorter_Compatible_Flag.

9. Dim_Shift (Workforce Shift Schedule Dimension)

Primary Key: `Shift_Key` (INT) | **Natural Key:** `Shift_Code` (VARCHAR) | **SCD:** Type 0 (Static)
Explicit Hierarchies: Shift: Shift Name → Shift Category (Day vs. Night Dispatch)
Key Attributes: `Shift_Key`, `Shift_Code`, `Shift_Name` (Morning Peak Dispatch, Evening Standard, Line-Haul Overnight), `Scheduled_Start_Time`, `Scheduled_End_Time`, `Standard_Shift_Hours` (e.g., 8.0), `Break_Duration_Minutes`, `Shift_Premium_Multiplier`.
Justification: Crucial for computing Deliveries per Rider-Hour without storing numeric measures in `Fact_Rider_Route_Assignment`.

4. Enterprise Dimensional Bus Matrix

The **Kimball Enterprise Bus Matrix** displays which conformed enterprise dimensions are shared across the three operational business processes, establishing the technical foundation for cross-process **drill-across** reporting.

Business Process (Fact Table)	Fact Type	Date	Hub	Route	Rider	Cust	Srv	Stat	Pkg	Shi
Last-Mile Delivery Attempts `Fact_Delivery_Attempt`	**Transaction**	**X** *(Attempt)*	**X** *(Origin)*	**X**	**X**	**X**	**X**	**X**	**X**	
Daily Hub Parcel Backlog `Fact_Daily_Hub_Backlog`	**Periodic Snapshot**	**X** *(Snapshot)*	**X**						**X**	
Rider Route Allocation `Fact_Rider_Route_Assignment`	**Factless**	**X** *(Assign)*	**X**	**X**	**X**					**X**

4.1 Conformed Dimension Cross-Analysis

- **Core Network Conformance (`Dim_Date`, `Dim_Hub`):** Shared across all three fact tables. Allows management to correlate daily sorting backlogs at a specific hub with delivery lateness and driver dispatch coverage for any calendar date.
- **Workforce Conformance (`Date`, `Rider`, `Route`):** Shared between `Fact_Delivery_Attempt` and `Fact_Rider_Route_Assignment`. Enables multi-fact drill-across queries to benchmark actual delivered packages against scheduled coverage rosters.
- **Physical Flow Conformance (`Date`, `Hub`, `Parcel_Type`):** Shared between `Fact_Delivery_Attempt` and `Fact_Daily_Hub_Backlog`. Exposes correlations between high-dwell parcel categories in sorting centers and recurring last-mile delivery failures on the road.

5. Star Schema Design vs. Snowflake Normalization

5.1 Why Snowflake Schemas Were Evaluated and Avoided

In a normalized snowflake schema, dimension hierarchies are decomposed into third normal form (3NF) relational sub-tables:

- `Dim_Route` → normalized out into `Dim_Zone` and `Dim_City`.
- `Dim_Hub` → normalized out into `Dim_City`, `Dim_Region`, and `Dim_Country`.
- `Dim_Customer` → normalized out into `Dim_Customer_Tier` and `Dim_Postal_Code`.

Architectural Rationale for Choosing Star Schema:

1. **Eliminating Relational Joins for Performance:** An analytical query aggregating delivery performance across geographic regions in a snowflake schema requires 4 sequential joins before reaching the fact table. In our star schema, region is denormalized directly inside `Dim_Hub` and `Dim_Route`, enabling single-join lookups and drastically reducing database CPU execution time.
2. **BI Usability & Self-Service Simplicity:** Business analysts using tools like PowerBI, Looker, or Tableau work much faster with simple, flat dimensions. Snowflaked webs of normalized tables confuse users and produce erroneous SQL outer-join results.
3. **Modern Columnar Storage Efficiency:** Analytical databases (e.g., Snowflake, BigQuery, ClickHouse) compress repetitive text columns (e.g., repeating `Region_Name = 'North'` across 500 rows) using dictionary and run-length encoding. The disk savings of 3NF snowflake normalization are negligible compared to the severe query performance penalty.

6. Slowly Changing Dimension (SCD) Policy

Operational business master data evolves over time. The LogiTrack data warehouse implements strict SCD rules to prevent retroactive corruption of historical reports:

Dimension Name	SCD Policy	Tracked Attributes	Implementation & Historical Reporting Behavior
`Dim_Rider`	**SCD Type 2**	`Home_Hub_Code`, `Vehicle_Type`, `Employment_Type`	When a rider transfers from Hub A to Hub B, the existing row is marked expired (`Current_Row_Indicator = 'Expired'`, `Expiration_Date = Current_Date`). A new row is inserted with a new surrogate `Rider_Key` and `Current_Row_Indicator = 'Current'`. Historical delivery transactions remain linked to the past Hub A record, preserving historical performance attribution.

Dim_Customer	**SCD Type 2**	Contract_Tier (Gold, Silver, Bronze), Billing_Street_Address	When a corporate client is upgraded from Silver to Gold tier, a new surrogate key is generated. Deliveries completed under the prior Silver contract remain attributed to the Silver tier, while new orders link to the Gold tier record.
Dim_Hub	**SCD Type 1**	Hub_Manager_Name, Contact_Phone, Street_Address	In-place overwrite. Corrections or administrative manager reassignments do not alter past sorting volumes and do not require historical tracking.
Dim_Route	**SCD Type 1**	Target_Daily_Stops, Average_Route_Distance_Km	In-place overwrite. Changes represent administrative corridor re-calibrations.
Dim_Date / Dim_Shift	**SCD Type 0**	All attributes	Static dimension tables. Values are immutable and pre-populated during warehouse setup.